Standard Operating Procedure for Resolving Critical QoS Congestion Event (ALARM_6852) on Ericsson PE Node WFD-PE-02 at Whitefield POP

Node: WFD-PE-02 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_6852

Description

The QoS congestion event (ALARM_6852) on Ericsson PE Node WFD-PE-02 at Whitefield POP is a critical failure domain that occurs due to microbursts, interface tail drops, and QoS queue starvation. This event is primarily caused by the PE node's inability to effectively manage and regulate traffic flow, leading to resource exhaustion and congestion on the network. The Ericsson PE node's role in QoS policy enforcement and traffic shaping makes it a critical component in maintaining network performance. The PE node uses the Resource Admission Control (RAC) mechanism to control resource allocation and admission control, which involves policing, shaping, and marking of traffic flows. However, in the event of microbursts or sudden increases in traffic volume, the PE node may struggle to keep up with the demand, leading to congestion and QoS queue starvation. Furthermore, the PE node's inability to effectively manage its interface resources, including buffers and queues, can exacerbate the congestion problem. To resolve this critical event, it is essential to identify and address the root cause of the congestion, which may involve reconfiguring QoS policies, adjusting resource admission control parameters, or optimizing interface resources.

Pre-requisite Checks

- Verify that the Whitefield POP is accessible and remote hands are available for assistance.
- Check the status of redundant links to ensure that they are operational and functioning correctly.
- Verify that the PE node's power supply and cooling systems are functioning properly.

Potential Root Causes

Cause	Probability
Microbursts or sudden increases in traffic volume	High
Interface resource exhaustion (buffer and queue overflow)	Medium
Ineffective QoS policy enforcement and traffic shaping	Low

Diagnostic Commands & Expected Outputs

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Command: show resource admission control status, Expected Output: RAC status: Active Policy name: QoS_policy_1 Priority: 1 , Description: This command displays the current RAC status, including the policy name, priority, and other relevant information. By analyzing this output, we can determine if the RAC mechanism is functioning correctly and if there are any issues with policy enforcement and traffic
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shaping.

Command: show interface statistics, Expected Output: Interface name: GigabitEthernet0/0/0 Input bytes: 1000000 Output bytes: 500000 , Description: This command displays the interface statistics, including input and output bytes. By analyzing this output, we can determine if there are any issues with interface resource utilization, including buffer and queue overflow.
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Resolution Steps

Step 1: Step: Reconfigure QoS policy to prioritize critical traffic flows, Command: config qos policy QoS_policy_1 priority 2, Description: This command reconfigures the QoS policy to prioritize critical traffic flows, which should help to alleviate congestion and QoS queue starvation.

Step 2: Step: Adjust resource admission control parameters, Command: config resource admission control parameters policer 1000 500, Description: This command adjusts the resource admission control parameters to limit the amount of traffic that can be admitted to the network, which should help to prevent microbursts and sudden increases in traffic volume.

Step 3: Step: Optimize interface resources, Command: config interface GigabitEthernet0/0/0 buffer-size 2000, Description: This command optimizes the interface resources by increasing the buffer size, which should help to prevent interface resource exhaustion and buffer overflow.

Post-Check Verification

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Command: show resource admission control status, Expected Output: RAC status: Active Policy name: QoS_policy_1 Priority: 2 , Description: This command verifies that the RAC mechanism is functioning correctly and that the QoS policy is being enforced correctly.

Command: show interface statistics, Expected Output: Interface name: GigabitEthernet0/0/0 Input bytes: 1000000 Output bytes: 500000 , Description: This command verifies that the interface resources are being utilized correctly and that there are no issues with buffer and queue overflow.
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Roll-back Procedures

- Step: Revert QoS policy to previous configuration, Command: config qos policy QoS_policy_1 priority 1, Description: This command reverts the QoS policy to its previous configuration, which should help to restore network performance.

- Step: Revert resource admission control parameters to previous values, Command: config resource admission control parameters policer 1000 500, Description: This command reverts the resource admission control parameters to their previous values, which should help to restore network performance.

- Step: Revert interface resources to previous configuration, Command: config interface GigabitEthernet0/0/0 buffer-size 1000, Description: This command reverts the interface resources to their previous configuration, which should help to restore network performance.